

India's Districts, Reclustered

What six years of NFHS-5 survey data reveal when 706 Indian districts are grouped by health, not by state.

706

districts clustered
across 36 states & UTs

How the six clusters were built

01

80+ NFHS-5 indicators

per district — nutrition, maternal & child health, sanitation, literacy, lifestyle risk

02

Cleaned & de-correlated

columns over 30% missing dropped; redundant pairs (correlation > 0.90) removed

03

Standardized, then K-Means

features scaled and grouped into $k = 6$ clusters (elbow method)

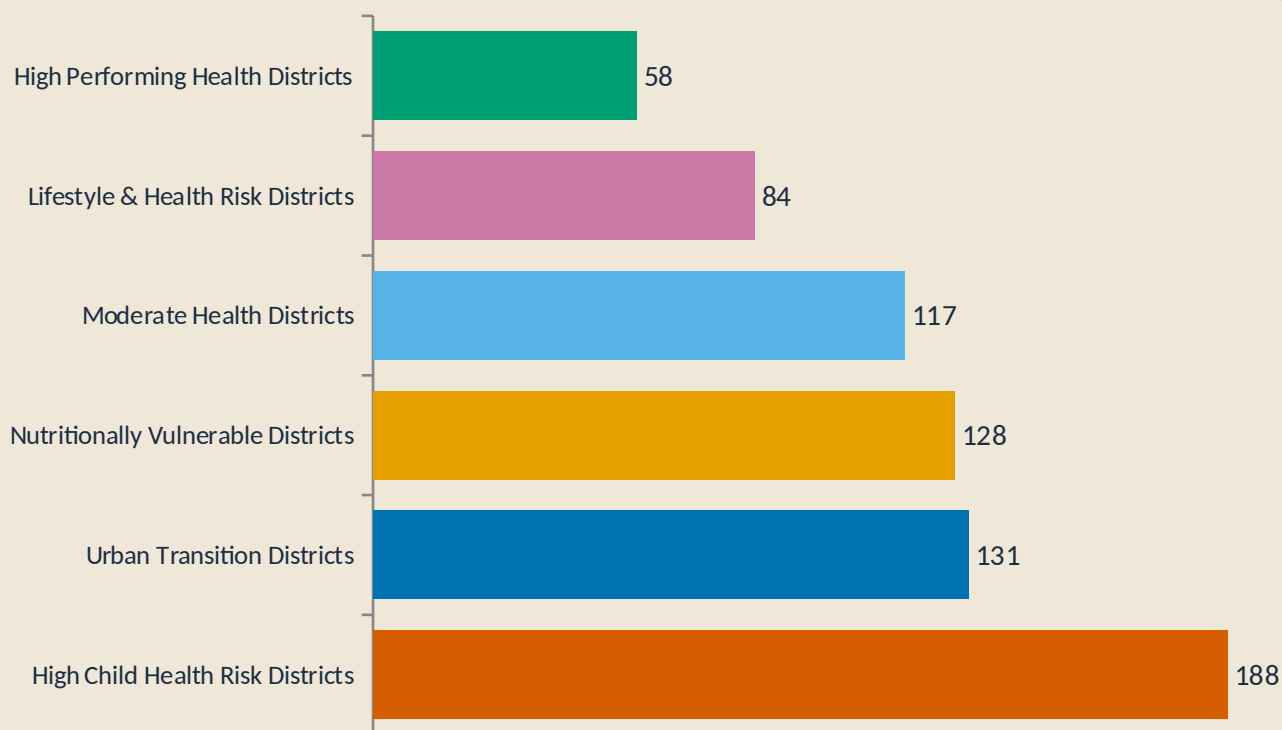
04

PCA for the map

two components reduce the feature space for visualization

Six health profiles, not two

706 districts sorted by K-Means into six distinct profiles — from best-performing to highest-risk.



Largest: High Child Health Risk (188 districts, 27%) · Smallest: High Performing Health (58, 8%)



Good health outcomes, less health infrastructure

Districts across Arunachal Pradesh, Meghalaya, Nagaland, Manipur & Mizoram cluster together as "High Performing Health Districts" — with the lowest anaemia of any group, despite the lowest rate of institutional births.

37.5%

Women (15-49) anaemic — the lowest of all six clusters, vs. 58-67% elsewhere. Institutional births here average just 68.7%, also the lowest.



Development doesn't erase health risk

Tamil Nadu, Telangana, Karnataka, Kerala & Andhra Pradesh dominate the "Urban Transition" cluster — near-universal institutional births and electricity access, but the highest lifestyle-disease burden in the country.

34.5%

Women overweight or obese — highest of any cluster, alongside the highest rate of high blood sugar (18.5%). Institutional births: 98.4%.



Where child health risk concentrates

Madhya Pradesh, Gujarat, Rajasthan, Maharashtra & Chhattisgarh make up the largest cluster in the country — 188 districts, over a quarter of India — flagged for child health risk.

72.6%

Children aged 6-59 months who are anaemic — the highest average of any cluster, and the single largest group of districts nationwide.



The stunting & underweight belt

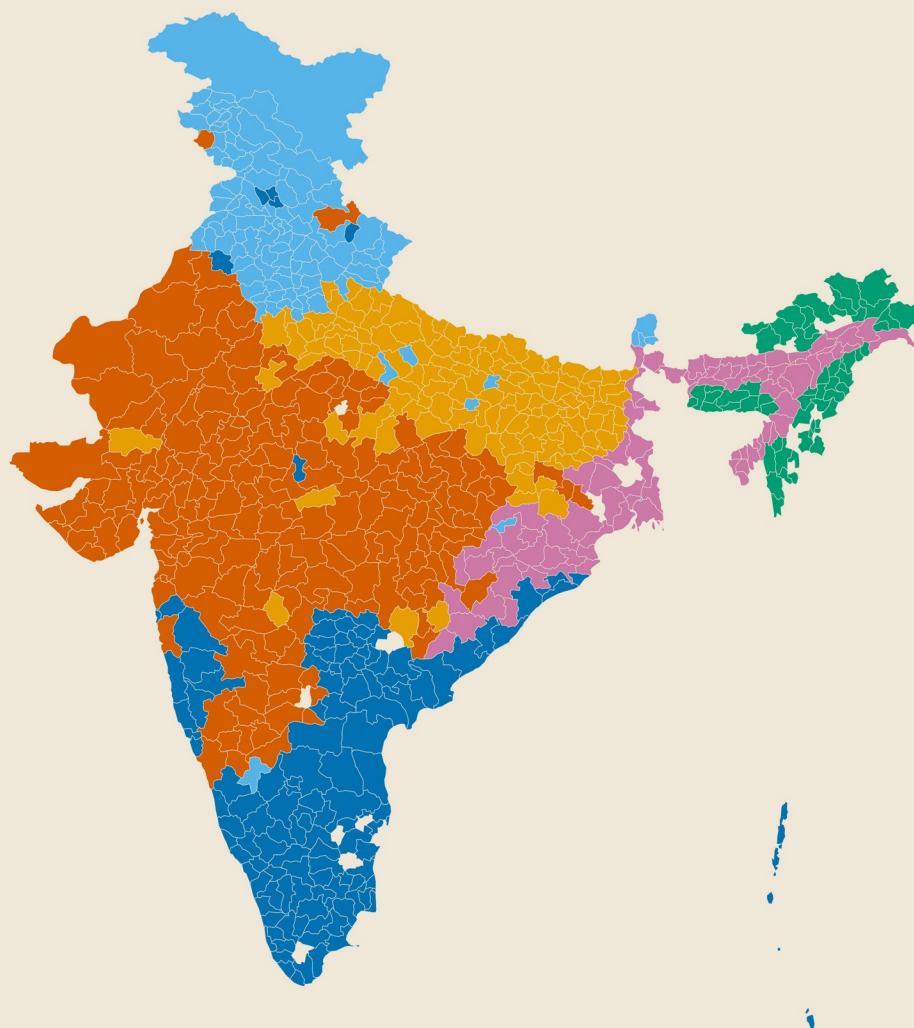
Uttar Pradesh, Bihar & Jharkhand anchor the "Nutritionally Vulnerable" cluster — the lowest female literacy of any group, paired with the country's highest child malnutrition rates.

41.7%

Children under five who are stunted — highest of the six clusters. Underweight prevalence: 37.1%. Women's literacy: 62.3%, the lowest average.

THE MAP

Same country, six different Indias



● High Performing Health

● Urban Transition

● Nutritionally Vulnerable

NFHS-5, 2019-21

● Moderate Health

● Lifestyle & Health Risk

● High Child Health Risk

One country. Six health realities.

State boundaries hide as much as they reveal. A district-level lens shows that health challenges in India cluster by lived conditions, not just by state lines — which is also where targeted policy has to aim.

Full interactive district map + analysis code available on request.